PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
1	Data Access & Integration	The solution must ingest and integrate a cohort dataset of 5,000 patient records containing demographic data (Age 15–79), clinical indicators (Blood Pressure, Glucose, BMI, Cholesterol, Chest Pain, Dizziness), and lifestyle behavioral metrics (Smoking status for 2,538 smokers, Alcohol consumption, Exercise Levels 0–2, Fatigue conditions).
2	Data Processing	The solution must clean, transform, and categorize raw clinical records into structured analytical dimensions: binning patient ages into cohorts (Teenager, Young Adult, Adult, Middle Age 45–59, Senior 60–74, Old 75–79), categorizing BMI index into risk tiers (Obese, Overweight, Normal, Underweight), and normalizing exercise frequencies (Level 0: 32.9%, Level 1: 33.5%, Level 2: 33.5%).
3	Business Analysis	The solution must execute healthcare risk algorithms to calculate key population health metrics: Overall Disease Prevalence Rates for Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%); Heart Disease Rate by BMI (Obese: 39.98%, Overweight: 32.91%, Normal: 25.97%, Underweight: 20.54%); Heart Disease Rate by Smoking Status (Smokers: 31.2% vs Non-smokers: 28.7%); and Age Group Heatmap Disease Rates (Old/Senior 100% Diabetes/Heart Disease, Middle Age 21.8% Diabetes / 11.7% Heart Disease).
4	Business Insights & Analytical Outputs	The solution must render an interactive 8-view Tableau Visual Story ('Story 1') comprising: (1) Disease Prevalence Comparison bar chart, (2) Patient Age Trend line chart (peak at age 50 bin), (3) BMI vs Heart Disease bar chart, (4) Age Group Bubble Chart (highlighting 60-74 and 75-79 risk bubbles), (5) Age Disease Heatmap matrix, (6) Clinical Scatter Plot (Blood Pressure 0–7000 vs Glucose 70–200 grouped by Heart Disease scale 5–20), (7) Exercise-level Pie Chart, and (8) Smoking Status comparative bar chart.
5	User Interaction & Analysis	The solution must enable interactive filtering and exploration for clinical officers (Dr. Michael Carter), analysts (Emily Rodriguez), and researchers (Dr. Priya Nair), allowing dynamic filtering across Measure Names, Age Bins, BMI Categories, Smoking Status, Exercise Levels, and real-time hover tooltips for patient risk stratification.
6	Data Update & Reporting	The solution must support automated visual dashboard updates when

		underlying EHR/EMR datasets are refreshed, providing exportable executive summary reports, high-risk patient flags, and public health policy story presentations.

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Tableau dashboards must feature an intuitive, user-friendly interface with clear color coding (e.g., green for low risk, dark red/orange for high risk), responsive filters, clear typography, and self-explanatory tooltips so healthcare stakeholders can interpret patient risk in under 3 seconds.
NFR-2	**Security**	The solution must comply with HIPAA and healthcare data privacy standards by anonymizing patient personal identifiable information (PII), enforcing role-based access control (RBAC) for clinical officers, analysts, and researchers, and securing data transmissions via SSL/TLS encryption.
NFR-3	**Reliability**	The data pipeline and visualization engine must ensure 99.9% data accuracy across all aggregations, ensuring zero calculation discrepancies between raw patient records and displayed KPI prevalence rates (Diabetes 34.92%, Heart Disease 29.98%, Stroke 24.86%).
NFR-4	**Performance**	Tableau story points and visual charts must load within 2 seconds upon initial request, and dynamic visual filters (e.g., filtering by smoking status or age bin) must update across all 8 story worksheets in under 1.5 seconds.
NFR-5	**Availability**	The solution hosted on Tableau Public / Server infrastructure must maintain 99.9% operational uptime, providing continuous 24/7 web access for healthcare professionals, clinical decision makers, and public health researchers.
NFR-6	**Scalability**	The architecture must be designed to seamlessly scale from the baseline 5,000 patient cohort to over 500,000 patient records without degradation in dashboard rendering performance or data query latency.